

TEST PLAN

Mobile Trading Application — iOS & Android

Version 1.0 | Pre-Release (T-14 Days)

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1. Overview

1.1 Objective

Validate that the mobile trading application is stable, secure, functionally correct, and ready for public release. Given that real user funds are involved and this is the app's first public release, this plan prioritises financial transaction integrity, authentication security, and platform stability above all else.

1.2 Document Scope

This document covers:

- Test Plan — strategy, scope, approach, schedule, and test cases
- Release Readiness Checklist — mandatory criteria before go/no-go decision
- Risk Matrix — identified risks, likelihood, impact, and mitigations

2. Scope

2.1 In Scope

- User registration and onboarding
- Authentication (login, logout, MFA, biometric)
- Account management
- Deposits and withdrawals
- Trade execution (buy/sell orders)
- Portfolio and balance updates
- Transaction history
- Market data display
- Push, email, and SMS notifications
- Error handling and recovery
- Security validation
- Performance and stability
- Cross-platform testing (iOS and Android)

2.2 Out of Scope

- Low-priority UI enhancements and cosmetic issues
- Nice-to-have features not included in the release scope
- Admin/back-office portals
- Third-party vendor/broker/exchange system testing (black-box boundary only)

2.3 Test Environments

Environment	Platform	Devices / OS Versions
Production Staging	iOS	iPhone 15 (iOS 17), iPhone 13 (iOS 16)
	Android	Samsung Galaxy S24 (Android 14), Pixel 7 (Android 13)
API / Backend	Staging API	Matches production configuration
Network	All	Wi-Fi, 4G/LTE, simulated 3G, airplane mode recovery

3. Test Strategy

Given the two-week horizon, zero existing coverage, and real user funds at risk, the strategy follows a strict risk-first prioritisation. Critical financial and authentication flows are tested before any other area.

3.1 Testing Types & Priority

#	Testing Type	Focus Areas	Priority	Week
1	Risk-Based Testing	User funds, financial transactions, authentication, data integrity, regulatory compliance	Critical	1–2
2	Smoke Testing	App launch, login, portfolio load, place trade, deposit/withdraw, logout	Critical	1
3	Functional Testing	Positive/negative scenarios, boundary values, error messages, business rules	High	1–2
4	Security Testing	Auth, session timeout, biometric, encryption, API security, log exposure	Critical	1
5	Cross-Platform Testing	iOS (latest + n-1), Android (latest + n-1), multiple screen sizes	High	1–2
6	API Testing	All financial endpoints, authentication APIs, error responses, edge cases	Critical	1
7	Performance Testing	App launch time, trade submission latency, market data refresh, network recovery	High	2
8	Manual Exploratory	Unscripted session-based testing of critical flows and edge cases	High	1–2
9	Regression Testing	Lightweight suite covering all business-critical flows; executed before release	Critical	2

3.2 Two-Week Test Schedule

Day(s)	Activity	Owner
Day 1	Environment setup, test data preparation, smoke test execution on both platforms	QA Lead
Day 2–3	Authentication, session, and security test execution	QA Engineer
Day 3–5	Financial flows: deposit, withdrawal, buy/sell, portfolio validation	QA Engineer
Day 5–7	API testing, error handling, edge cases, network interruption scenarios	QA Engineer
Day 7–9	Cross-platform validation (iOS/Android), screen sizes, device rotation	QA Engineer
Day 9–11	Performance and stability testing, exploratory sessions	QA Lead
Day 11–12	Defect retesting and verification of fixes	QA Engineer
Day 13	Full regression run on both platforms	QA Lead
Day 14	Go/No-Go decision, test summary report, release recommendation	QA Lead + PM

4. Test Cases

TC ID	Area	Test Scenario	Test Steps	Expected Result	Priority
TC001	Auth	User registration	Register with valid name, email, password, phone	Account created; verification email sent	High
TC002	Auth	Registration — invalid fields	Submit form with missing/invalid data	Field-level validation messages displayed; form not submitted	High
TC003	Auth	Login — valid credentials	Enter correct username and password	User authenticated and lands on home screen	Critical
TC004	Auth	Login — invalid credentials	Enter incorrect username or password	Login denied; appropriate error shown; no account lockout on first attempt	Critical
TC005	Auth	MFA authentication	Complete login using OTP/authenticator app	User authenticated only after valid MFA code	Critical
TC006	Auth	Biometric login	Log in using Face ID or fingerprint	Authentication succeeds with registered biometric	High
TC007	Auth	Logout	Tap logout	Session ends; redirected to login; session token invalidated	High
TC008	Auth	Session timeout	Leave app idle beyond timeout threshold	User automatically logged out and prompted to re-authenticate	High
TC009	Trading	View portfolio	Open portfolio screen after login	All holdings, quantities, and balances display correctly	Critical
TC010	Trading	View market data	Search and open a tradeable instrument	Current price, bid/ask, and instrument details shown accurately	High
TC011	Trading	Buy — valid order	Purchase a valid quantity of XRP with sufficient funds	Order executes; confirmation shown; portfolio and balance update	Critical
TC012	Trading	Sell — valid order	Sell held XRP	Order executes; portfolio and balance update correctly	Critical
TC013	Trading	Buy — insufficient funds	Attempt to buy with balance below order value	Order rejected; clear error message; no balance deducted	Critical
TC014	Trading	Sell — insufficient holdings	Attempt to sell more BTC than owned	Order prevented; appropriate error displayed	Critical
TC015	Trading	Trade confirmation screen	Submit a buy or sell order	Confirmation screen shows correct instrument, quantity, price, and fees	Critical

TC ID	Area	Test Scenario	Test Steps	Expected Result	Priority
TC016	Trading	Portfolio update post-trade	Complete a successful trade	Holdings and account balance reflect trade within expected latency	Critical
TC017	Trading	Duplicate order prevention	Submit the same order twice rapidly (double-tap)	Only one order is placed; duplicate prevented	Critical
TC018	Funds	Deposit funds	Initiate and complete a valid deposit	Balance updates correctly; transaction appears in history	Critical
TC019	Funds	Withdraw funds	Withdraw available funds below balance	Withdrawal succeeds; balance reduces; transaction history updated	Critical
TC020	Funds	Over-withdrawal	Attempt to withdraw more than available balance	Transaction rejected; clear error shown; balance unchanged	Critical
TC021	Funds	Transaction history	Open transaction history	All past transactions shown accurately with correct amounts, dates, and status	High
TC022	Security	Sensitive data in logs	Run a trade and inspect device logs	No account numbers, tokens, passwords, or financial data in logs	Critical
TC023	Security	Screenshot prevention	Attempt to screenshot authentication or fund screens	Screenshots blocked or blurred on sensitive screens	Critical
TC024	Security	API — auth token validation	Call a financial API with an expired/invalid token	API returns 401; no data returned	Critical
TC025	Security	Brute-force lockout	Enter incorrect password 5+ times consecutively	Account temporarily locked; user notified	High
TC026	Network	Network interruption during trade	Disconnect network mid trade submission	User receives clear error; no duplicate trade created	Critical
TC027	Network	App relaunch after kill	Force-close and reopen the app	Session state handles correctly per security policy (re-auth or resume)	High
TC028	Network	Background/foreground resume	Send app to background and return after 30+ seconds	App resumes correctly; no data loss or unexpected logout	Medium
TC029	Network	Network recovery	Toggle airplane mode during market data load	App recovers and refreshes data on reconnect	High
TC030	Performance	App launch time	Cold-start the app and time until home screen loads	App launches in under 3 seconds on supported devices	High
TC031	Performance	Trade submission latency	Submit a market order and measure response time	Order confirmation received within 2 seconds under normal network	High

TC ID	Area	Test Scenario	Test Steps	Expected Result	Priority
TC032	Performance	Market data refresh	Observe live price updates on the trading screen	Price refreshes at expected interval without freezing or crashing	High
TC033	Cross-Platform	iOS smoke suite	Execute all Critical TCs on iOS 17 and iOS 16	All critical flows pass on both iOS versions	Critical
TC034	Cross-Platform	Android smoke suite	Execute all Critical TCs on Android 14 and Android 13	All critical flows pass on both Android versions	Critical
TC035	Cross-Platform	Device rotation	Rotate device during trade entry and portfolio view	UI adapts without data loss or layout breakage	Medium
TC036	Notifications	Trade confirmation push	Complete a trade with push notifications enabled	Push notification received within 30 seconds of trade execution	Medium
TC037	Notifications	Price alert notification	Set a price alert and wait for trigger	Notification delivered accurately when price threshold is hit	Low
TC038	Error Handling	Backend failure — trade	Simulate API 500 error on trade submission	User shown a clear error; app remains stable; no partial state	High
TC039	Error Handling	API error handling	Simulate backend failure on various endpoints	User receives meaningful error messages; app does not crash	High
TC040	Error Handling	Input boundary values	Enter 0, negative numbers, and max values in trade quantity and amount fields	Appropriate validation applied at all boundaries	High

5. Release Readiness Checklist

All CRITICAL items below must be checked before any go/no-go decision. HIGH items must be resolved or have accepted mitigations. MEDIUM items may ship with documented known issues.

#	Category	Requirement	Severity	Status	Notes
1	Functional	All Critical test cases have passed on iOS and Android	Critical	☐ Pending	
2	Functional	Zero open Critical or High-severity defects	Critical	☐ Pending	
3	Functional	Trade execution (buy/sell) validated end-to-end with real test funds	Critical	☐ Pending	
4	Functional	Deposit and withdrawal flows verified with correct balance updates	Critical	☐ Pending	
5	Functional	Portfolio and balance display accurately after all trade types	Critical	☐ Pending	
6	Functional	Duplicate trade prevention confirmed under rapid submission	Critical	☐ Pending	
7	Functional	Transaction history shows all transactions accurately	High	☐ Pending	
8	Security	No sensitive data (tokens, passwords, account numbers) in device logs	Critical	☐ Pending	
9	Security	Screenshot prevention active on auth and financial screens	Critical	☐ Pending	
10	Security	API authentication token validation confirmed (expired tokens rejected)	Critical	☐ Pending	
11	Security	Session timeout enforced and tested on both platforms	High	☐ Pending	
12	Security	MFA enforced and validated for all login flows	Critical	☐ Pending	
13	Security	Biometric authentication working correctly on supported devices	High	☐ Pending	
14	Security	Brute-force lockout confirmed after threshold exceeded	High	☐ Pending	
15	Security	All API calls use HTTPS/TLS; no plain-text communication	Critical	☐ Pending	
16	Financial	Over-withdrawal rejected with correct error; balance unchanged	Critical	☐ Pending	
17	Financial	Insufficient funds error prevents trade from executing	Critical	☐ Pending	
18	Financial	Insufficient holdings error prevents oversell	Critical	☐ Pending	
19	Financial	All financial calculations (fees, totals, balances) verified for accuracy	Critical	☐ Pending	
20	Performance	App cold-start completes in < 3 seconds on supported devices	High	☐ Pending	
21	Performance	Trade order confirmed within 2 seconds under normal network conditions	High	☐ Pending	
22	Performance	No crashes or ANR errors during full smoke test execution	Critical	☐ Pending	
23	Performance	App recovers correctly after network interruption during trade	Critical	☐ Pending	
24	Platform	Smoke suite passed on iOS latest and n-1 versions	Critical	☐ Pending	
25	Platform	Smoke suite passed on Android latest and n-1 versions	Critical	☐ Pending	
26	Platform	App store review submitted and approved (iOS App Store / Google Play)	Critical	☐ Pending	

#	Category	Requirement	Severity	Status	Notes
27	Platform	App builds tested on real devices (not emulator-only)	High	☐ Pending	
28	Compliance	Privacy policy and terms of service accessible within the app	High	☐ Pending	
29	Compliance	Error messages do not expose internal system information	High	☐ Pending	
30	Compliance	Production monitoring/alerting configured and validated	High	☐ Pending	
31	Compliance	Rollback plan confirmed and tested with the engineering team	Critical	☐ Pending	
32	Compliance	On-call escalation plan in place for day-one production issues	High	☐ Pending	
33	Documentation	Test summary report completed and signed off	High	☐ Pending	
34	Documentation	All known issues documented with severity and workaround	High	☐ Pending	
35	Documentation	Go/No-Go decision recorded with stakeholder sign-off	Critical	☐ Pending	

6. Risk Matrix

Risks are scored on Likelihood (1–5) × Impact (1–5).

Score ≥ 15 = Critical action required.

Score 8–14 = High priority mitigation.

Score 4–7 = Medium.

Score < 4 = Low / Accept.

Risk ID	Risk Description	Category	Likelihood (1–5)	Impact (1–5)	Score	Rating	Mitigation
R01	No existing test suite — unknown defect density	Process	4	5	20	Critical	Immediate risk-based test execution starting with financial flows; all-hands on testing Week 1
R02	Financial transaction bugs leading to incorrect fund movement	Financial	3	5	15	Critical	Mandatory positive and negative testing of all deposit, withdrawal, and trade flows with test funds
R03	Duplicate trade execution (double-tap or network retry)	Financial	4	5	20	Critical	Test idempotency controls; confirm backend deduplication logic; test rapid repeat submissions
R04	Authentication bypass or session token exposure	Security	3	5	15	Critical	Full security test pass on auth flows; verify token expiry, API auth headers, and log sanitisation
R05	Sensitive financial data exposed in device logs or screenshots	Security	3	5	15	Critical	Inspect logs post-test; verify screenshot blocking on sensitive screens
R06	App crashes or freezes on supported iOS or Android versions	Stability	4	4	16	Critical	Smoke test on all target OS versions Day 1; crash analytics integration before release
R07	Network interruption during trade causes partial or duplicate order	Reliability	4	4	16	Critical	Test all network interruption scenarios; verify backend handles retries safely
R08	Balance or portfolio not updated after trade execution	Financial	3	5	15	Critical	Verify real-time balance sync after every trade type; test with delay simulation
R09	Performance degradation under concurrent users at launch	Performance	3	4	12	High	Load test key endpoints; define acceptable latency thresholds before release
R10	MFA or biometric authentication failure on specific devices	Auth	3	4	12	High	Test on minimum 4 real devices across iOS and Android; document known device limitations
R11	App store approval rejection delaying release	Platform	2	5	10	High	Submit builds for review by Day 7 to allow time for resubmission if rejected
R12	Market data displaying stale or incorrect prices	Data	3	4	12	High	Validate data feed integration; test refresh intervals and data source failover
R13	Push notifications delayed or not delivered	Notifications	3	3	9	High	Test notification delivery on both platforms; define acceptable SLA

Risk ID	Risk Description	Category	Likelihood (1–5)	Impact (1–5)	Score	Rating	Mitigation
R14	UI layout broken on specific screen sizes or resolutions	UI	3	3	9	High	Test on minimum 4 screen sizes; prioritise most common device configurations
R15	Insufficient test time leading to gaps in coverage	Process	4	4	16	Critical	Strict risk-based prioritisation; daily progress check; escalate scope concerns to PM immediately
R16	Dev team ships new code during test window introducing regressions	Process	4	4	16	Critical	Code freeze from Day 10; any exceptions require QA sign-off before merge
R17	No rollback plan if critical defect found post-launch	Operations	3	5	15	Critical	Confirm rollback plan with engineering before Day 14 go/no-go
R18	Regulatory or compliance failure (missing disclosures, terms)	Compliance	2	5	10	High	Verify all required legal content accessible in-app before release

7. Deliverables

Deliverable	Description	Due
Test Plan (this document)	Scope, strategy, schedule, test cases, risk matrix	Day 1
Smoke Test Report	Pass/fail results for all Critical TCs on both platforms	Day 2
Defect Log (Jira)	All bugs logged with severity, steps to reproduce, and evidence	Ongoing
Security Test Report	Summary of all security validation results	Day 7
Performance Test Report	Latency, stability, and network recovery results	Day 11
Regression Test Report	Full regression run results for both platforms	Day 13
Test Summary Report	Aggregate coverage, defect summary, open risks	Day 13
Release Recommendation	Go/No-Go recommendation with rationale	Day 14

8. Entry & Exit Criteria

8.1 Entry Criteria (Before Testing Begins)

- Test environment is configured and accessible on both platforms
- Test accounts and test funds are provisioned
- Application build is stable enough to pass a basic app launch check
- Access to Jira or defect tracking tool is confirmed
- This test plan has been reviewed by QA Lead

8.2 Exit Criteria (Before Release)

- All 35 Release Readiness Checklist items are marked complete or have accepted exceptions
- Zero open Critical defects
- All High defects resolved or accepted with documented mitigation
- Full regression suite passed on iOS and Android
- Test Summary Report and Release Recommendation signed off by QA Lead and Product Manager

9. Assumptions & Dependencies

9.1 Assumptions

- A stable build will be available by Day 1
- Test accounts with real test funds can be provisioned for financial flow testing
- The dev team will implement a code freeze no later than Day 10
- App store submissions will be initiated no later than Day 7 to allow time for review

9.2 Dependencies

- Backend API staging environment must be production-equivalent
- Push notification service must be active in staging
- Biometric testing requires real physical devices (not emulators)
- Any third-party integrations (payment rails, brokerage APIs) must be available in staging

10. Approvals & Sign-Off

Role	Name	Signature	Date
QA Lead			
Product Manager			
Engineering Lead			
CTO / Release Owner			